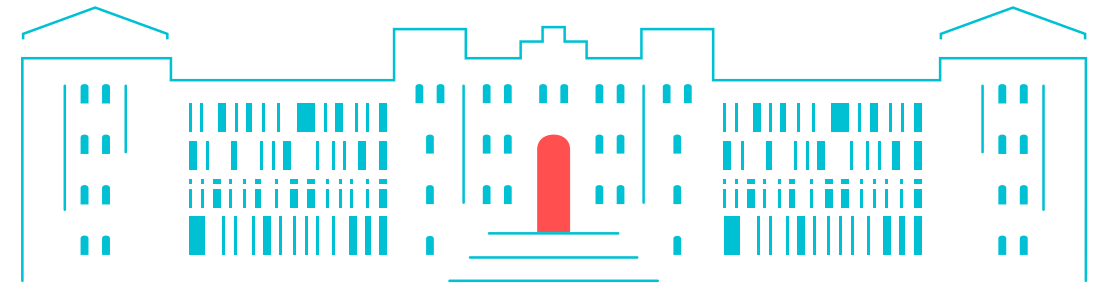
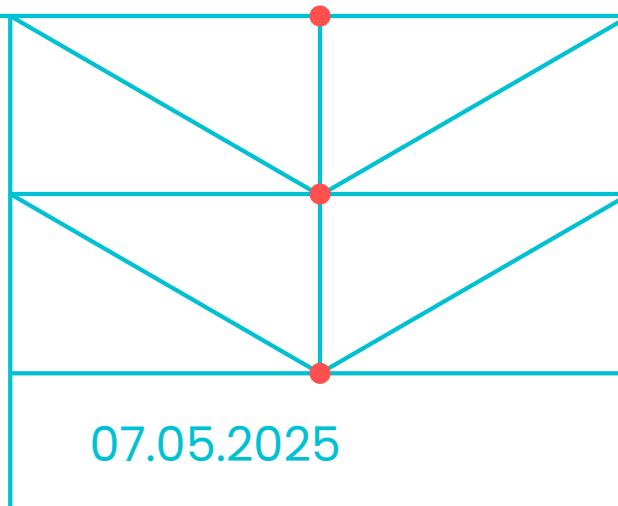


Parallelization of Solvers

Fynn Bense

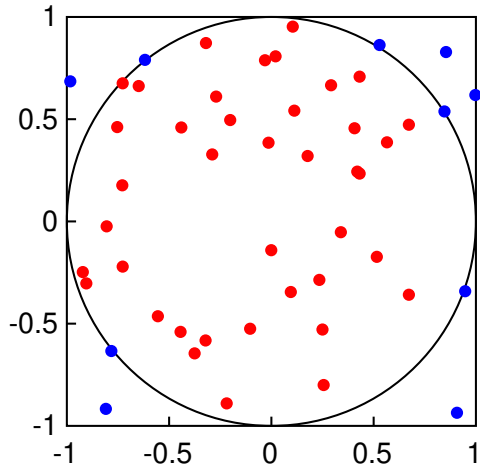
Chair Computational Mathematics
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University of
Technology

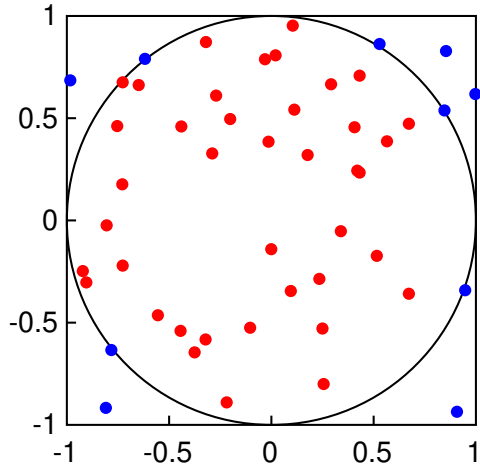


06 – Matrix Vector MPI

- Recap of basic MPI
- MPI-parallel Monte Carlo for computation of π
- Matrix-Vector product
- Parallel computation of rows
- Parallel distribution of column



Note that the data exchange at the end is fairly complicated.
This can be achieved much easier using collective operations.



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Homework

- Take the `mc_mpi_pi_reduce.c` and let a process other than zero report the result when using `MPI_REDUCE` or `MPI_ALLREDUCE` and see how you get different results

Reduce

```
int MPI_Reduce(const void *sendbuf, void *recvbuf, int count, MPI_Datatype datatype, MPI_Op op, int root, MPI_Comm comm)
```

The `sendbuf` parameter is an array of elements of type `datatype` that each process wants to reduce. The `recvbuf` is only relevant on the process with a rank of `root`. The `recvbuf` array contains the reduced result and has a size of `sizeof(datatype) * count`. The `op` parameter is the operation that you wish to apply to your data.

Gather

```
int MPI_Reduce(const void *sendbuf, int sendcount, MPI_Datatype sendtype, void *recvbuf, int recvcount, MPI_Datatype recvttype, int root, MPI_Comm comm)
```

The `sendbuf` parameter is an array of `sendcount` elements of type `sendtype` that each process sends. The `recvbuf` array contains the gathered results only on process with rank equal to `root` and has a size of `sizeof(recvttype) * recvcount * WOLRD_SIZE`.

Task

Given $A \in \mathbb{R}^{n_{rows} \times n_{cols}}$ and $x \in \mathbb{R}^{n_{cols}}$, calculate

$$Ax = b \in \mathbb{R}^{n_{rows}}$$

Small example

$$Ax = \begin{pmatrix} 1 & 2 & 3 & 0 \\ 6 & 0 & 1 & 2 \\ 1 & 1 & 0 & 0 \\ 2 & 0 & 0 & 0 \end{pmatrix} \begin{pmatrix} 1 \\ 1 \\ 1 \\ 1 \end{pmatrix} = \begin{pmatrix} 6 \\ 9 \\ 2 \\ 2 \end{pmatrix} = b$$

How to distribute this to $P = 2$ processes? We need to calculate

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How to distribute this to $P = 2$ processes? We need to calculate

$$b_i = \sum_{j=0}^{N-1} a_{i,j} x_j$$

For $P0$ and $P1$ we distribute:

$$\begin{pmatrix} 1 & 2 & 3 & 0 \\ 6 & 0 & 1 & 2 \\ 1 & 1 & 0 & 0 \\ 2 & 0 & 0 & 0 \end{pmatrix} \begin{pmatrix} 1 \\ 1 \\ 1 \\ 1 \end{pmatrix} = \begin{pmatrix} 6 \\ 9 \\ 2 \\ 2 \end{pmatrix}$$

Distribution

Rows per process: $n_r = \frac{4}{2} = 2$

Rows for two processes: $R(0) = \{0, 2\}, R(1) = \{1, 3\}$ (Other distributions possible)

Question 1

For P processes, how to compute $R(r)$ for $r \in \{0, \dots, P-1\}$?

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$$n_r = \frac{n_{\text{rows}}}{P}$$

$$R(r) = \{r, r + P, r + 2 \cdot P, \dots, r + (n_r - 1) \cdot P\}$$

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Rows per process: $n_r = \frac{4}{2} = 2$

Rows for two processes: $R(0) = \{0, 2\}$, $R(1) = \{1, 3\}$ (Other distributions possible)

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$$R(r) = \{r, r + P, r + 2 \cdot P, \dots, r + (n_r - 1) \cdot P\}$$

Question 2

For $k \in \{0, \dots, n_{rows} - 1\}$, how to find r such that $k \in R(r)$?

$$k = r + iP$$

$$r = k \% P$$

Task

Implement row distributed MPI version, starting from `matrix_vector_row.c`

For $P0$ and $P1$ we distribute:

$$\begin{pmatrix} 1 & 2 & 3 & 0 \\ 6 & 0 & 1 & 2 \\ 1 & 1 & 0 & 0 \\ 2 & 0 & 0 & 0 \end{pmatrix} \begin{pmatrix} 1 \\ 1 \\ 1 \\ 1 \end{pmatrix} = \begin{pmatrix} 6 \\ 9 \\ 2 \\ 2 \end{pmatrix}$$

Distribution

Columns per process: $n_c = \frac{4}{2} = 2$

Columns for two processes: $C(0) = \{0, 1\}$, $C(1) = \{2, 3\}$ (Other distributions possible)

Computation

$$b_i = \sum_{j=0}^{n_{cols}-1} a_{i,j}x_j = \sum_{r=0}^{P-1} \sum_{j \in C(r)} a_{i,j}x_j$$

Matrix vector product

1. Each process needs to read all entries of A and x it owns
2. Each process computes for $i = 0, \dots, n_{\text{rows}} - 1$:

$$b_i^{(r)} := \sum_{j \in C(r)} a_{i,j} x_j$$

3. For $i = 0, \dots, n_{\text{rows}} - 1$ compute:

$$b_i = \sum_{r=0}^{P-1} b_i^{(r)}$$

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$$b_i = \sum_{r=0}^{P-1} b_i^{(r)}$$

Homework

Implement this (column distributed MPI) for the example A and x where all entries are 1 (large example).